

Extraction of high quality RNA from *Gardnerella* species

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Method Article

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Abstract

This protocol describes an approach for the purification of high quality RNA from *Gardnerella* species based on a commercially available kit (PureLink RNA Mini Kit, Invitrogen). Initial RNA extraction following the manufacturer's protocol resulted in poor RNA yield and quality from *Gardnerella* spp. based on A_{260}/A_{280} measurements and RNA integrity number (RIN). Modifications made to the manufacturer's recommended protocol including enzymatic lysis with lysozyme and proteinase K resulted in significantly higher RNA yield (≥ 25 ng/mL) and RNA quality ($A_{260}/A_{280} \sim 2.0$, $RIN \geq 8.0$). RNA extracts resulting from this protocol are appropriate for use in gene expression studies based on real-time quantitative PCR or transcriptomics.

Introduction

Real-time quantitative PCR is a powerful tool for targeted study of bacterial gene expression. The MIQE (Minimum Information for Publication of Quantitative Real-Time PCR Experiments) guidelines were created to provide standards for ensuring the integrity of the scientific literature, including guidelines for RNA quality and integrity (1). Given the huge diversity of bacterial cell composition, achieving high quality RNA extracts for downstream analysis requires the optimization of extraction protocols. Detailed protocols for the purification of high quality RNA from *Gardnerella* spp. have not been reported. To optimize an RNA purification protocol for *Gardnerella* spp. we initially evaluated multiple commercially available RNA preparation kits using the manufacturers' protocols and assessed the results in terms of yield and quality. The PureLink RNA Mini Kit (Invitrogen)(2) provided the highest quality results in this initial comparison so we focused our efforts of optimizing this kit for *Gardnerella*. Addition of mutanolysin and proteinase K to the lysozyme solution, increased volume of lysis buffer and extended incubation time at 37°C during lysis resulted in significant improvements. The final protocol presented here consistently resulted in high quality RNA, indicated by an $A_{260}/A_{280} \sim 2.0$ and $RIN \geq 8.0$. We have successfully used RNA from this protocol in RT-qPCR and transcriptome sequencing experiments.

Reagents

RNase AWAY Surface Decontaminant (Thermo Fisher Scientific, 21-236-21)

PureLink RNA Mini Kit (Invitrogen, 12183018a) (Contains Lysis buffer*, Wash buffer I*, Wash buffer II, RNase free water, spin cartridges, collection tubes, recovery tubes) [* Lysis Buffer and Wash Buffer I contain guanidine isothiocyanate (an irritant). This chemical is harmful when in contact with the skin, or when it is inhaled or ingested.]

PureLink DNase Set (Invitrogen, 12185010)

10% SDS (Invitrogen, AM9822)

0.1 mM EDTA (Invitrogen, AM9912)

Lysozyme (Thermo Scientific, 89833)

Proteinase K Solution (20 mg/mL), RNA grade (Invitrogen, 25530049)

Mutanolysin (Sigma-Aldrich, M9901)

Tris (1 M), pH 8.0, RNase-free (Invitrogen, AM9855G)

100% ethanol

2-mercaptoethanol

RNA ScreenTape Ladder (Agilent, 5067-5578)

RNA ScreenTape Sample Buffer (Agilent, 5067-5577)

RNA ScreenTape (Agilent, 5067-5576)

Equipment

1cc syringes (25G x 5/8" needle) (Thermo Fisher Scientific, 14-817-125)

Pipettes (P1000, P200, P20, P10)

RNase-free microcentrifuge tubes

Microcentrifuge set to 4°C

Microcentrifuge set to 25°C

Water bath at 37°C

Heating block at 72°C

Ice

TapeStation (Agilent)

NanoDrop Microvolume Spectrophotometer (Thermo Fisher Scientific)

Procedure

Before you start:

NOTE: This protocol is designed to process cell pellets containing $\leq 1 \times 10^9$ bacteria each. The procedure presented here starts with fresh bacterial cell culture pellets. *In our experience, pellets harvested by*

centrifugation and stored at -80°C can be used successfully for RNA extraction.

RNA extraction should be performed in a RNase free area. This can be achieved by wiping the pipets, microcentrifuge and the work bench using RNase AWAY, and using RNase free consumables and reagents.

Set microcentrifuge to 4°C.

If analyzing RNA concentration and integrity by TapeStation, thaw RNA ScreenTape Ladder on ice and allow Sample Buffer and RNA ScreenTape to equilibrate at room temperature for 30 minutes prior to using. Preheat heating block to 72 °C.

Prepare solutions:

Lysis Buffer with 2-mercaptoethanol

Prepare 700 µL of freshly prepared Lysis Buffer with 1% (v/v) 2-mercaptoethanol for each cell pellet ($\leq 1 \times 10^9$ bacteria) to be processed.

Add 10 µL of 2-mercaptoethanol for each 1 mL Lysis Buffer needed.

Wash Buffer II with Ethanol

Add 16 mL 96–100% ethanol to Wash Buffer II (provided in the kit). Check the box on the Wash Buffer II label to indicate that ethanol was added. Store Wash Buffer II with ethanol at room temperature.

Lysozyme Solution

Prepare 200 µL per sample, containing:

2 µL 1M Tris pH 8.0, RNase-free

185.2 µL 0.1M EDTA

0.8 µL lysozyme (250 mg/mL, in 10 mM Tris pH 8.0, RNase-free)

2 µL mutanolysin (10,000 U/mL)

10 µL proteinase K (20 mg/mL)

PureLink DNase Mixture

Prepare 80 μL per sample containing:

8 μL 10X DNase I Reaction Buffer

10 μL Resuspended DNase ($\sim 3 \text{ U}/\mu\text{L}$)

62 μL RNase Free Water

RNA Extraction:

1. Harvest $\leq 1 \times 10^9$ log-phase bacteria and transfer them to an appropriately sized microcentrifuge tube. Keep microcentrifuge tubes on ice while harvesting all samples. Centrifuge at $2000 \times g$ for 5 minutes at 4°C to pellet cells. Discard the supernatant.
2. Set microcentrifuge to 25°C .
3. Add 200 μL of prepared lysozyme solution to the cell pellet and resuspend by pipetting up and down. Keep all samples on ice while adding lysozyme solution and vortex all microcentrifuge tubes for 10-15 seconds.
4. Add 0.5 μL 10% SDS solution to all samples while on ice. Vortex to mix well.
5. Incubate lysate for 15 minutes in 37°C water bath.
6. Add 700 μL Lysis Buffer prepared with 2-mercaptoethanol. Vortex for 10-15 seconds.
7. Homogenize the cell lysate by passing the lysate 5 times through a 25G x 5/8" needle attached to a syringe.
8. Centrifuge the lysate at $12,000 \times g$ for 2 minutes at 25°C , then transfer supernatant to a clean RNase-free microcentrifuge tube.
9. Add 250 μL 100% ethanol to the supernatant recovered in Step 8.
10. Mix thoroughly by vortexing to disperse any visible precipitate that may form after adding ethanol.
11. Transfer 700 μL of sample (including any remaining precipitate) to a Spin Cartridge (with a Collection Tube).

12. Centrifuge both Spin Cartridge and Collection tube at 12,000 x *g* for 15 seconds at 25°C. Discard the flow-through and insert the Spin Cartridge in the same Collection Tube.
13. Repeat steps 11-12 until remaining homogenate has been passed through the column.
14. Add 350 µL Wash Buffer I to the Spin Cartridge containing the bound RNA. Centrifuge at 12,000 x *g* for 15 seconds at 25°C. Discard flow through and the Collection Tube. Insert the Spin Cartridge into a new Collection Tube.
15. Add 80 µL PureLink DNase mixture directly onto the surface of the Spin Cartridge membrane.
16. Incubate at room temperature (or in microcentrifuge set at 25°C) for 15 minutes.
17. Add 350 µL Wash Buffer I to the Spin Cartridge. Centrifuge at 12,000 x *g* for 15 seconds at 25°C. Discard flow through and reinsert the Spin Cartridge into the same Collection tube.
18. Add 500 µL of Wash Buffer II with ethanol to the Spin Cartridge.
19. Centrifuge at 12,000 x *g* for 15 seconds at 25°C. Discard flow through and reinsert the Spin Cartridge into the same Collection tube.
20. Repeat steps 18-19, once.
21. Centrifuge the Spin Cartridge at 12,000 x *g* for 1 minute at 25°C to dry the membrane with bound RNA. Discard the Collection Tube and reinsert the Spin Cartridge into a Recovery Tube.
22. Add 30 µL – 100 µL RNase-Free Water (30 µL will maximize RNA yield) to the center of the Spin Cartridge.
23. Incubate at room temperature (or in microcentrifuge set at 25°C) for 2 minutes.
24. Centrifuge Spin Cartridge and Recovery Tube at 12,000 x *g* for 1 minute at 25°C.
25. Immediately place samples on ice.

Quality control of extracted RNA:

1. RNA ScreenTape Assay (Agilent) – RNA Integrity Number and concentration
2. Nanodrop for 260/280 ratio

Store the extracted RNA at -80°C.

Troubleshooting

Time Taken

It takes approximately 2 hours to process 6 samples in parallel, starting from frozen cell pellets. Once the protocol is started, it should be completed in its entirety and the RNA extracts stored at -80°C.

Anticipated Results

RNA extractions from 1.6 mL broth cultures of *G. swidsinskii* harvested in exponential phase (OD₆₀₀ 0.3-0.6) yielded 1.5-4.5 µg of RNA (30 µL of ~50-150 ng/µL) per sample with RIN > 7.5 (**Table 1**). Additional examples of typical TapeStation results for successful and unsuccessful extractions are shown in **Figure 1**.

References

1. Bustin, S. A., Benes, V., Garson, J. A., Hellemans, J., Huggett, J., Kubista, M., Mueller, R., Nolan, T., Pfaffl, M. W., Shipley, G. L., Vandesompele, J., & Wittwer, C. T. The MIQE guidelines: minimum information for publication of quantitative real-time PCR experiments. *Clinical Chemistry*, 55(4), 611–622 (2009).
2. PureLink RNA Mini Kit, User Guide. Life Technologies, Publication Number MAN0000406 Revision: 14 June 2012

Figures

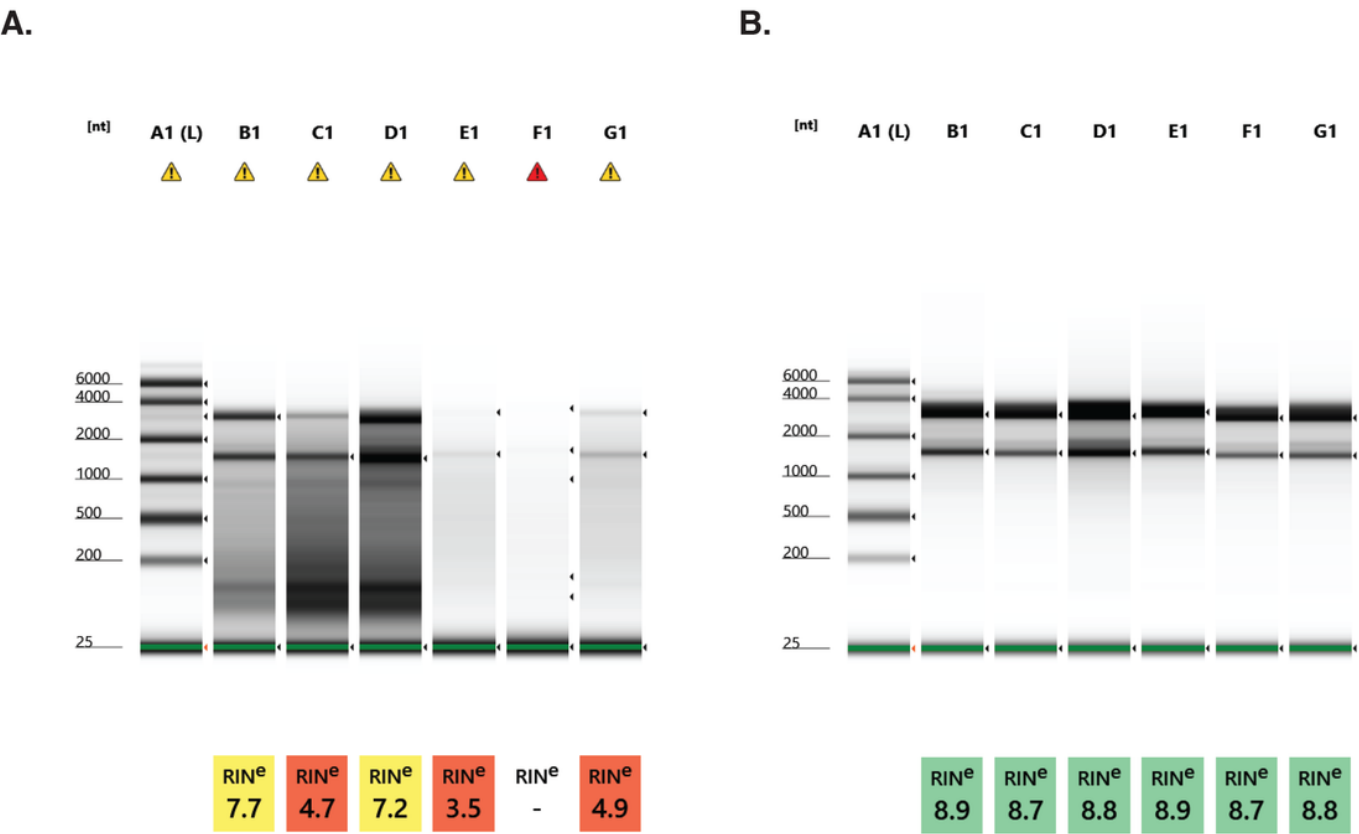


Figure 1. TapeStation results for extractions resulting in (A) poor quality, degraded RNA and (B) high quality RNA with RIN >8. The poor results shown in (A) could be the result of RNase activity, improper handling temperatures during the extraction, etc.

Figure 1

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Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [Table1.docx](#)